SG No. 125 D_{4h}^3 $P4/nbm$ [tetragonal]

* plus set: +[0, 0, 0]

Table 1: Wyckoff site: 2a, site symmetry: 422

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[1,2,3,4,5,6,7,8]
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[9,10,11,12,13,14,15,16]

Table 2: Wyckoff site: 2b, site symmetry: 422

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[9,10,11,12,13,14,15,16]

Table 3: Wyckoff site: 2c, site symmetry: -42m

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[1,2,5,6,11,12,15,16]
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[3,4,7,8,9,10,13,14]

Table 4: Wyckoff site: 2d, site symmetry: -42m

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[1,2,5,6,11,12,15,16]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[3,4,7,8,9,10,13,14]

Table 5: Wyckoff site: 4e, site symmetry: ..2/m

No.	position	mapping
1	[0, 0, 0]	[1,7,9,15]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,8,10,16]
3	$\left[\frac{1}{2}, 0, 0\right]$	[3,5,11,13]
4	$\left[0, \frac{1}{2}, 0\right]$	[4,6,12,14]

Table 6: Wyckoff site: 4f, site symmetry: ..2/m

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,7,9,15]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,8,10,16]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,5,11,13]
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[4,6,12,14]

Table 7: Wyckoff site: `4g`, site symmetry: `4..`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[5,6,7,8]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[9,10,11,12]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[13,14,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,2,15,16]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[3,4,13,14]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[7,8,9,10]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	`[2,8]`
3	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[3,5]`
4	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	`[4,6]`
5	$[-x,\ -x,\ 0]$	`[9,15]`
6	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[10,16]`
7	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[11,13]`
8	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	`[12,14]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	`[1,7]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[2,8]`
3	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	`[3,5]`
4	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[4,6]`
5	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[9,15]`
6	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[10,16]`
7	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[11,13]`
8	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[12,14]`

Table 11: Wyckoff site: `8k`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	`[1,6]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	`[2,5]`

continued ...

Table 11

No.	position	mapping
3	$\left[\frac{1}{4},\, x,\, 0\right]$	[3,7]
4	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, 0\right]$	[4,8]
5	$\left[-x,\, \frac{3}{4},\, 0\right]$	[9,14]
6	$\left[x+\frac{1}{2},\, \frac{3}{4},\, 0\right]$	[10,13]
7	$\left[\frac{3}{4},\, -x,\, 0\right]$	[11,15]
8	$\left[\frac{3}{4},\, x+\frac{1}{2},\, 0\right]$	[12,16]

Table 12: Wyckoff site: 8l, site symmetry: .2.

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, \frac{1}{2}\right]$	[1,6]
2	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{2}\right]$	[2,5]
3	$\left[\frac{1}{4},\, x,\, \frac{1}{2}\right]$	[3,7]
4	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[4,8]
5	$\left[-x,\, \frac{3}{4},\, \frac{1}{2}\right]$	[9,14]
6	$\left[x+\frac{1}{2},\, \frac{3}{4},\, \frac{1}{2}\right]$	[10,13]
7	$\left[\frac{3}{4},\, -x,\, \frac{1}{2}\right]$	[11,15]
8	$\left[\frac{3}{4},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[12,16]

Table 13: Wyckoff site: 8m, site symmetry: ..m

No.	position	mapping
1	$[x,\, -x,\, z]$	[1,15]
2	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, z\right]$	[2,16]
3	$\left[x+\frac{1}{2},\, x,\, z\right]$	[3,13]
4	$\left[-x,\, \frac{1}{2}-x,\, z\right]$	[4,14]
5	$\left[\frac{1}{2}-x,\, -x,\, -z\right]$	[5,11]
6	$\left[x,\, x+\frac{1}{2},\, -z\right]$	[6,12]
7	$[-x,\, x,\, -z]$	[7,9]
8	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$	[8,10]

Table 14: Wyckoff site: 16n, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[2]
3	$\left[\frac{1}{2}-y,\, x,\, z\right]$	[3]
4	$\left[y,\, \frac{1}{2}-x,\, z\right]$	[4]
5	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[5]
6	$\left[x,\, \frac{1}{2}-y,\, -z\right]$	[6]
7	$[y,\, x,\, -z]$	[7]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[8]
9	$[-x,\, -y,\, -z]$	[9]

continued ...

Table 14

No.	position	mapping
10	$\left[x+\frac{1}{2}, y+\frac{1}{2}, -z\right]$	[10]
11	$\left[y+\frac{1}{2}, -x, -z\right]$	[11]
12	$\left[-y, x+\frac{1}{2}, -z\right]$	[12]
13	$\left[x+\frac{1}{2}, -y, z\right]$	[13]
14	$\left[-x, y+\frac{1}{2}, z\right]$	[14]
15	$[-y, -x, z]$	[15]
16	$\left[y+\frac{1}{2}, x+\frac{1}{2}, z\right]$	[16]